

A Counterexample to the Corrected Banach Projection Problem

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Source Metadata

Source paper: K. Mahesh Krishna, *Paulsen and Projection Problems for Banach Spaces*, arXiv:2201.00991, 2022.

Target statement: Problem 2.14, “Projection problem for Banach spaces”, on pages 7–8 of the source PDF.

Open Problem Evidence

Here we state projection problem for Banach spaces.

Problem 2.14. (*Projection problem for Banach spaces*) *Let $\mathcal{X}$ be a d -dimensional Banach space with Auerbach basis $(\{\zeta_k\}_{k=1}^d, \{u_k\}_{k=1}^d)$ ($\{\zeta_k\}_{k=1}^d$ is the collection of dual functionals).*

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Find the function $g : (0, 1) \times \mathbb{N} \times \mathbb{N} \rightarrow [0, \infty)$ satisfying the following: If $P : \mathcal{X} \rightarrow \mathcal{X}$ is a projection (idempotent) of rank n satisfying

$$(1 - \varepsilon) \frac{n}{d} \leq \|Pu_k\|^2 = \|\zeta_k P\|^2 = |\zeta_k(Pu_k)| \leq (1 + \varepsilon) \frac{n}{d}, \quad \forall 1 \leq k \leq d,$$

then there exists a projection (idempotent) $Q : \mathcal{X} \rightarrow \mathcal{X}$ with

$$\|Qu_k\|^2 = \|\zeta_k Q\|^2 = |\zeta_k(Pu_k)| = \frac{n}{d}, \quad \forall 1 \leq k \leq d,$$

satisfying

$$\sum_{k=1}^d \frac{1}{2} (\|Pu_k - Qu_k\|^2 + \|\zeta_k P - \zeta_k Q\|^2) \leq g(\varepsilon, n, d).$$

We already remarked that Paulsen and projection problems were connected using chordal distance. Even though we are unable to connect Paulsen and projection problems for Banach spaces, we introduce the

Corrected Reading

As printed, the conclusion of Problem 2.14 asks for a projection Q such that

$$\|Qu_k\|^2 = \|\zeta_k Q\|^2 = |\zeta_k(Pu_k)| = \frac{n}{d} \quad (1 \leq k \leq d),$$

which still contains the original projection P in the diagonal scalar. The earlier lane packet records the resulting literal counterexample. This packet addresses the natural corrected reading

$$\|Qu_k\|^2 = \|\zeta_k Q\|^2 = |\zeta_k(Qu_k)| = \frac{n}{d} \quad (1 \leq k \leq d).$$

Idea of the Proof

We choose a two-dimensional norm for which one prescribed rank-one projection has the exact equalities required in the hypothesis, with diagonal values $3/5$ and $2/5$. These are within the allowed interval around $1/2$. However, an exactly flat corrected projection would force a unit vector whose two coordinate moduli are both $1/\sqrt{2}$. The chosen norm puts every such vector strictly inside the unit ball, so the corrected conclusion is impossible.

Theorem 1. *The corrected Banach projection problem obtained from Problem 2.14 of arXiv:2201.00991 by replacing $|\zeta_k(Pu_k)|$ with $|\zeta_k(Qu_k)|$ in the conclusion is false.*

Proof. Let

$$a = \sqrt{\frac{3}{5}}, \quad b = \sqrt{\frac{2}{5}}.$$

On $X = \mathbb{R}^2$, define

$$\|(x_1, x_2)\| = \max\{|x_1|, |x_2|, |ax_1 + bx_2|\}.$$

Let e_1, e_2 be the standard basis and let e_1^*, e_2^* be the coordinate functionals. Since the norm dominates $|x_1|$ and $|x_2|$, we have $\|e_k^*\| \leq 1$, and equality follows by evaluating at e_k . Also $\|e_k\| = 1$. Thus this is an Auerbach basis.

Put $u = (a, b)$ and $\phi(x_1, x_2) = ax_1 + bx_2$. Then

$$\|u\| = \max\{a, b, a^2 + b^2\} = 1, \quad \|\phi\| = 1, \quad \phi(u) = 1.$$

Define

$$Px = \phi(x)u.$$

Then $P^2 = P$ and $\text{rank } P = 1$. For $k = 1, 2$,

$$Pe_k = \phi(e_k)u, \quad e_k^*P = e_k^*(u)\phi.$$

Since $\|u\| = \|\phi\| = 1$, it follows that

$$\|Pe_1\|^2 = \|e_1^*P\|^2 = |e_1^*(Pe_1)| = a^2 = \frac{3}{5},$$

and

$$\|Pe_2\|^2 = \|e_2^*P\|^2 = |e_2^*(Pe_2)| = b^2 = \frac{2}{5}.$$

With $d = 2$, $n = 1$, and $\varepsilon = 1/5$, the required interval is

$$\left[(1 - \varepsilon)\frac{n}{d}, (1 + \varepsilon)\frac{n}{d}\right] = \left[\frac{2}{5}, \frac{3}{5}\right].$$

Therefore P satisfies the hypothesis of the corrected projection problem.

We now show that the corrected conclusion is impossible. Suppose, toward a contradiction, that a rank-one projection Q satisfies

$$\|Qe_k\|^2 = \|e_k^*Q\|^2 = |e_k^*(Qe_k)| = \frac{1}{2} \quad (k = 1, 2).$$

Write $Q = \psi \otimes v$, so $Qx = \psi(x)v$, where $v \neq 0$ and $\psi(v) = 1$. Put $\alpha_k = \psi(e_k)$ and $\beta_k = e_k^*(v)$. From $\|Qe_k\|^2 = 1/2$, we get

$$|\alpha_k| \|v\| = \frac{1}{\sqrt{2}}.$$

From $|e_k^*(Qe_k)| = 1/2$, we get

$$|\alpha_k \beta_k| = \frac{1}{2}.$$

Consequently, for both $k = 1, 2$,

$$|\beta_k| = \frac{\|v\|}{\sqrt{2}}.$$

Thus the normalized vector $w = v/\|v\|$ has coordinate moduli

$$|e_1^*(w)| = |e_2^*(w)| = \frac{1}{\sqrt{2}}.$$

For any choices of signs,

$$\left\| \left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}} \right) \right\| \leq \max \left\{ \frac{1}{\sqrt{2}}, \frac{a+b}{\sqrt{2}} \right\} = \frac{a+b}{\sqrt{2}} < 1,$$

because

$$(a+b)^2 = 1 + \frac{2\sqrt{6}}{5} < 2.$$

So every vector with coordinate moduli $1/\sqrt{2}$ has norm strictly less than 1. This contradicts $\|w\| = 1$. Hence no such Q exists, and the corrected projection problem is false. $\square$

Verification Notes

The included script `code/verify_polyhedral_counterexample.py` checks the common hypothesis values and the strict inequality for all four sign choices of vectors with coordinate moduli $1/\sqrt{2}$. The computation is only arithmetic support; the proof above is symbolic.

Command used:

```
python code/verify_polyhedral_counterexample.py
```

Novelty and Search Bounds

The run indexes were searched for arXiv:2201.00991, Paulsen problem, projection problem, approximate Schauder frame, and related phrases. No prior packet for this source was present before this lane's work. Web searches on 2026-06-28 for the exact title and projection-problem phrases returned the source arXiv record and no correction or later answer. This is a bounded novelty check, not an exhaustive literature review.

Limitations

This packet settles the Banach projection problem in the corrected universal form described above. It does not settle the separate Banach Paulsen problem for approximate Schauder frames, nor any restricted projection problem limited to Hilbert spaces or to special Banach spaces where exact flat projections may exist.

Human Review Recommendation

Review the corrected-statement interpretation and the rank-one obstruction in the two-dimensional norm

$$\|(x_1, x_2)\| = \max\{|x_1|, |x_2|, |ax_1 + bx_2|\}.$$

The decisive line is that exact flatness would force a normalized vector with coordinate moduli $1/\sqrt{2}$, while the chosen norm gives every such vector norm strictly less than 1.

References

- [1] K. Mahesh Krishna. *Paulsen and Projection Problems for Banach Spaces*. arXiv:2201.00991, 2022.